



FactoryOps

Predictive Maintenance & Factory Operations Platform

Intelligent monitoring, prediction, maintenance, and incident management for modern factory operations.

SMART OPERATIONS. STRONGER TOMORROW.

FactoryOps

Smart Operations. Stronger Tomorrow.

01 - PROBLEM

The Problem

Traditional factory maintenance is reactive, not predictive. Teams respond to failures after they occur, causing avoidable downtime and operational risk.

Unexpected Machine Failures

Unplanned breakdowns halt production lines and cause costly downtime across manufacturing units.

Manual Monitoring Gaps

Manual checks make it difficult to catch early warning signs before critical assets suffer severe damage.

Unusable Sensor Data

Raw sensor readings remain trapped in silos and are difficult to interpret without structured automated analysis.

No Centralized Visibility

Maintenance teams lack a unified real-time view of machine health, failure predictions, and open incidents.

FactoryOps Solution

FactoryOps brings critical factory operations into one centralized platform — from monitoring to maintenance tracking.

STEP 01

Monitor

Monitor machines and sensor conditions in real time across all lines.

KEY CAPABILITIES

- Real-time telemetry stream
- Multi-sensor data tracking
- Threshold anomaly alerts

STEP 02

Analyze

Interpret operational and sensor data for actionable intelligence.

KEY CAPABILITIES

- Automated data processing
- Pattern & trend evaluation
- Cross-sensor correlation

STEP 03

Predict

Identify machine health status and failure risk with AI scoring.

KEY CAPABILITIES

- Health score calculation
- Failure probability %
- Remaining days estimate

STEP 04

Maintain

Support maintenance decisions with data-backed recommendations.

KEY CAPABILITIES

- Prescriptive repair guidance
- Work order prioritization
- Root cause diagnostics

STEP 05

Track

Track incidents and operational issues floor-wide in real time.

KEY CAPABILITIES

- Centralized incident log
- Real-time status tracking
- Audit trail logging

FactoryOps Architecture

The platform follows a clean three-tier architecture: the Streamlit frontend delivers the user interface, the FastAPI backend handles all API services and data processing, and operational data from factory machines feeds the prediction models.

Users & Admins

Access the FactoryOps application interface for operational management

User Layer

Streamlit Frontend

Interactive application interface and visualization dashboards

UI Layer

FastAPI Backend

API services, analytical routing, and request handling infrastructure

API Layer

Factory Data Layer

Machines, real-time sensors, and AI prediction models

Data Layer

Application Workflow

FactoryOps provides a centralized workspace for accessing all main factory operations modules. Users authenticate before accessing the application.

Login & Access

Admin and User portals with authentication before accessing the application modules.

Dashboard

Overview of factory health, operational status, and active system alerts.

Machines

View machine inventory and search individual machine specifications easily.

Sensors

View sensor readings for individual machines across telemetry dimensions.

Predictions

View machine health, failure probability, and risk classification analytics.

Maintenance & Incidents

Access maintenance recommendations and track operational incidents seamlessly.

Factory Dashboard Metrics

The dashboard provides a centralized overview of factory health and operational status at a glance.

Factory Status

Overall factory operating status and live system health summary.

Total Machines

Total number of machines actively monitored in real time.

Machine Health

Healthy, Warning, and Critical machine breakdown distribution.

Average Health

Aggregated average machine health percentage score.

Availability

Current machine operational availability and uptime metric.

Active Alerts

Currently active operational and critical sensor alerts.

Maintenance Due

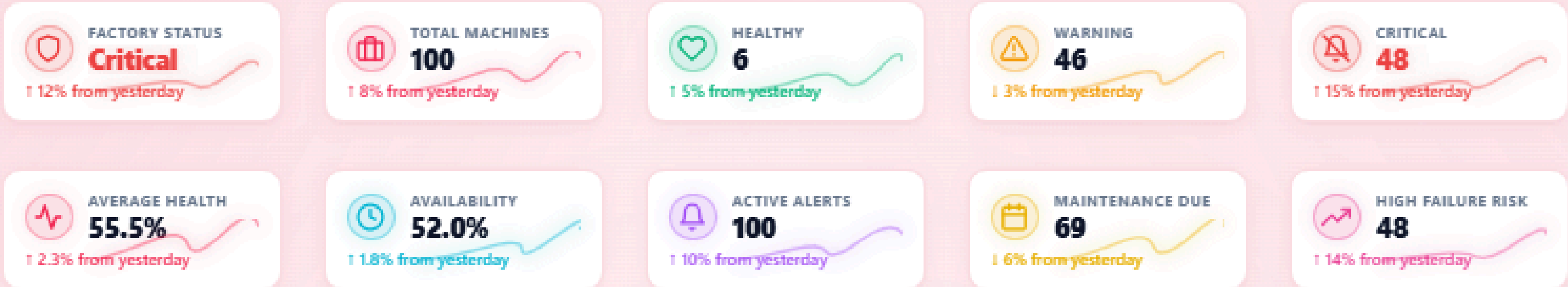
Machines requiring immediate or upcoming maintenance attention.

High Failure Risk

Machines with elevated predicted failure risk probabilities.

Factory Dashboard

Real-time overview of your factory operations and predictive insights



Health Overview

Overall equipment health distribution

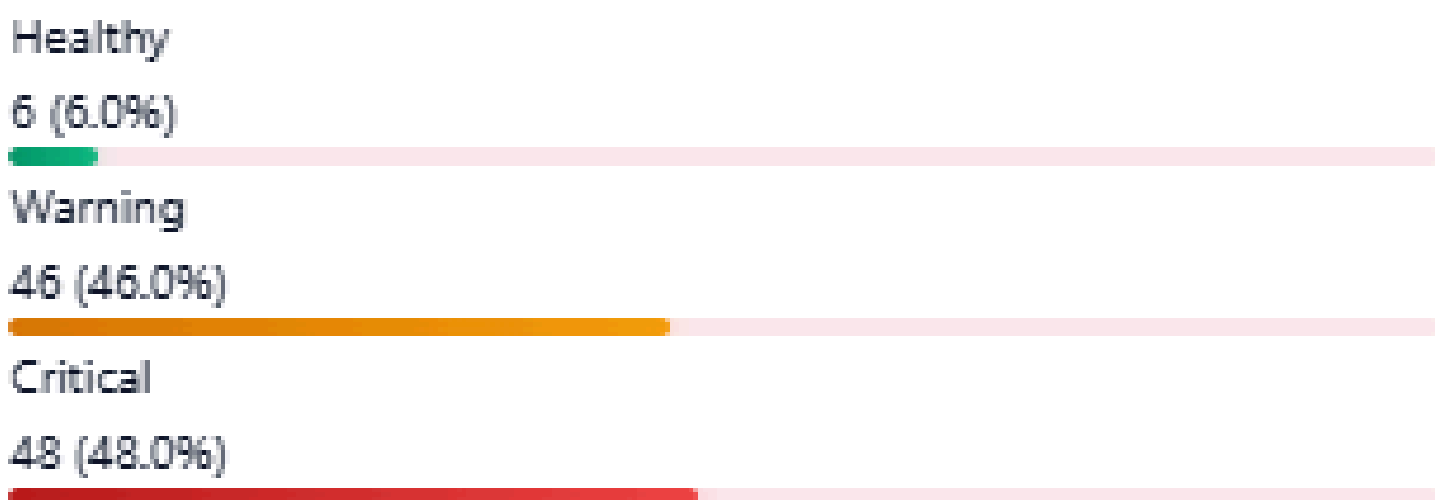
Machine Health Overview

Distribution of current machine health status



Healthy	6 (6.0%)
Warning	46 (46.0%)
Critical	48 (48.0%)

Health Summary



Quick Actions

View Machines

View Predictions

View Incidents

Machine & Sensor Monitoring

FactoryOps provides machine inventory information and sensor readings for monitoring operating conditions across the factory floor.

Machine Monitoring

- View complete machine inventory
- Search and filter individual machines
- Monitor machine-related operational info

Sensor Monitoring

- View sensor readings for individual machines
- Track real-time operating conditions

Sensor Parameters

Temperature Vibration Pressure
Humidity Voltage Current

Predictive Maintenance

FactoryOps presents machine health and failure-risk indicators to support maintenance decisions.

Machine Health Score

Indicates the current health condition of a machine based on telemetry analysis.

Failure Probability

Shows the predicted probability of machine failure in percentage values.

Risk Level

Categorizes machine failure risk into Healthy, Warning, or Critical levels.

Predicted Failure

Displays predicted failure information and root cause diagnostics.

Estimated Failure Days

Provides an estimated time and remaining useful days until potential failure.

Confidence Score

Indicates the confidence score associated with the AI prediction model.

Maintenance & Incident Management

FactoryOps connects machine risk information with maintenance activities and incident tracking to support operational decision-making.

Maintenance Management

- View maintenance records and complete historical service logs
- Access recommendations linked to machine predictions and AI risk scores
- Connect maintenance decisions directly to failure risk information
- Prioritize work orders based on estimated failure urgency and remaining days
- Reduce Mean Time to Repair (MTTR) with guided diagnostic recommendations
- Track technician sign-offs and schedule automated preventive routines

Incident Management

- View and filter reported operational incidents by active status
- Track operational issues and live critical alerts across factory units
- Monitor incident resolution progress across maintenance and engineering teams
- Automatically trigger alerts when telemetry thresholds exceed safety guidelines
- Assign emergency response tickets directly to available floor engineers
- Maintain comprehensive audit logs for compliance, safety, and root-cause analysis

Conclusion & Future Scope

FactoryOps provides a centralized platform for monitoring, prediction, maintenance planning, and incident tracking.

Key Project Takeaways

- Centralized real-time machine and sensor telemetry monitoring across production lines
- Sensor information aggregation with automated anomaly processing
- Comprehensive machine health visibility with live risk categorization
- AI-powered failure-risk analytics and estimated days-to-failure insights
- Actionable maintenance planning support paired with live incident tracking
- Maximized asset uptime and higher overall equipment effectiveness (OEE)

Future Roadmap & Enhancements

- Real-time high-frequency sensor streaming pipeline integration (MQTT / Kafka)
- Historical machine-health trend visualization with interactive time-series charts
- Granular role-based access control (RBAC) and enterprise authentication
- Automated ML model retraining, threshold tuning, and multi-channel instant alerts
- Deep ERP & CMMS integration for automated work order creation
- Mobile supervisor app with push notifications for critical floor emergencies

FactoryOps — Code Architecture

Clean modular separation across Frontend, Backend, API services, AI Logic, and Database models.

Frontend & Configuration

- `app.py` Streamlit application
- `.streamlit/` UI config
- `assets/` Images & resources
- `factoryops.db` SQLite database
- `.gitignore` Git configuration

Backend & API Layer

- `config.py` / `db_config.py`
- `dashboard_api` , `machine_api`
- `sensor_api` , `prediction_api`
- `maintenance_api` , `incident_api`

Services & AI Logic

- `main.py` / Health Monitoring
- Anomaly Detection
- Failure Prediction & Root Cause
- Maintenance Recommendations
- Dashboard Intelligence

Database & Models: `seed_database.py` Sample data • Machine, Sensor, Prediction, Maintenance & Incident Models

THANK YOUU

TEAM MEMBERS:

Premalatha N K

Mahesh Kashyap

Nekkala Jyotshna

Krishna Kanth

Sri Vatsavi

Manasa

Sanjana